Class 08: Breast Cancer Mini Project

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Table of contents

Background

The goal of this mini-project is for you to explore a complete analysis using the unsupervised learning techniques covered in class.

Today we will analyze a biopsy data-set from fine needle aspiration (FNA) of breast mass.

Data Import

The data is made available as a CSV file for download. We can read this in using `read.csv()` and store it as an object which I will call **"wisc.df"**.

```r
wisc.df <- read.csv("WisconsinCancer.csv", row.names = 1)

head( wisc.df, 4 )
```

```
         diagnosis radius_mean texture_mean perimeter_mean area_mean
842302           M       17.99        10.38         122.80    1001.0
842517           M       20.57        17.77         132.90    1326.0
84300903         M       19.69        21.25         130.00    1203.0
84348301         M       11.42        20.38          77.58     386.1
         smoothness_mean compactness_mean concavity_mean concave.points_mean
842302           0.11840          0.27760         0.3001             0.14710
842517           0.08474          0.07864         0.0869             0.07017
84300903         0.10960          0.15990         0.1974             0.12790
84348301         0.14250          0.28390         0.2414             0.10520
         symmetry_mean fractal_dimension_mean radius_se texture_se perimeter_se
842302          0.2419                0.07871    1.0950     0.9053        8.589
842517          0.1812                0.05667    0.5435     0.7339        3.398
84300903        0.2069                0.05999    0.7456     0.7869        4.585
84348301        0.2597                0.09744    0.4956     1.1560        3.445
         area_se smoothness_se compactness_se concavity_se concave.points_se
842302    153.40      0.006399        0.04904      0.05373           0.01587
842517     74.08      0.005225        0.01308      0.01860           0.01340
84300903   94.03      0.006150        0.04006      0.03832           0.02058
84348301   27.23      0.009110        0.07458      0.05661           0.01867
         symmetry_se fractal_dimension_se radius_worst texture_worst
842302       0.03003             0.006193        25.38         17.33
842517       0.01389             0.003532        24.99         23.41
84300903     0.02250             0.004571        23.57         25.53
84348301     0.05963             0.009208        14.91         26.50
         perimeter_worst area_worst smoothness_worst compactness_worst
842302            184.60     2019.0           0.1622            0.6656
842517            158.80     1956.0           0.1238            0.1866
84300903          152.50     1709.0           0.1444            0.4245
84348301           98.87      567.7           0.2098            0.8663
         concavity_worst concave.points_worst symmetry_worst
842302            0.7119               0.2654         0.4601
842517            0.2416               0.1860         0.2750
84300903          0.4504               0.2430         0.3613
84348301          0.6869               0.2575         0.6638
         fractal_dimension_worst
842302                   0.11890
842517                   0.08902
84300903                 0.08758
84348301                 0.17300
```

Make sure we exclude the `diagnosis` column from the data-set that we use for further analysis - this is the expert diagnosis as either M or B:

```r
wisc.data <- wisc.df[,-1]
diagnosis <- as.factor(wisc.df$diagnosis)
```

Eploratory Data Analysis

> Q1: How many observations are in this dataset?

```r
nrow(wisc.data)
```

```
[1] 569
```

> Q2: How many of the observations have a malignant diagnosis?

We can use a `table()` function to do this.

```r
table(wisc.df$diagnosis)
```

```

  B   M 
357 212 
```

> Q3: How many variables/features in the data are suffixed with `_mean`?

We can use the `grep()` function to help us. It is similar to a "ctrl + f" finder function. Keeping the `value = TRUE` returns the actual value that matches what you look for while leaving it as `value = FALSE` returns the integer value where it is located.

```r
length( grep( "_mean", colnames(wisc.data), value = TRUE) )
```

```
[1] 10
```

Principal Componenet Analysis (PCA)

The main function of `prcomp(x, scale = FALSE, center = FALSE)` only requires X.

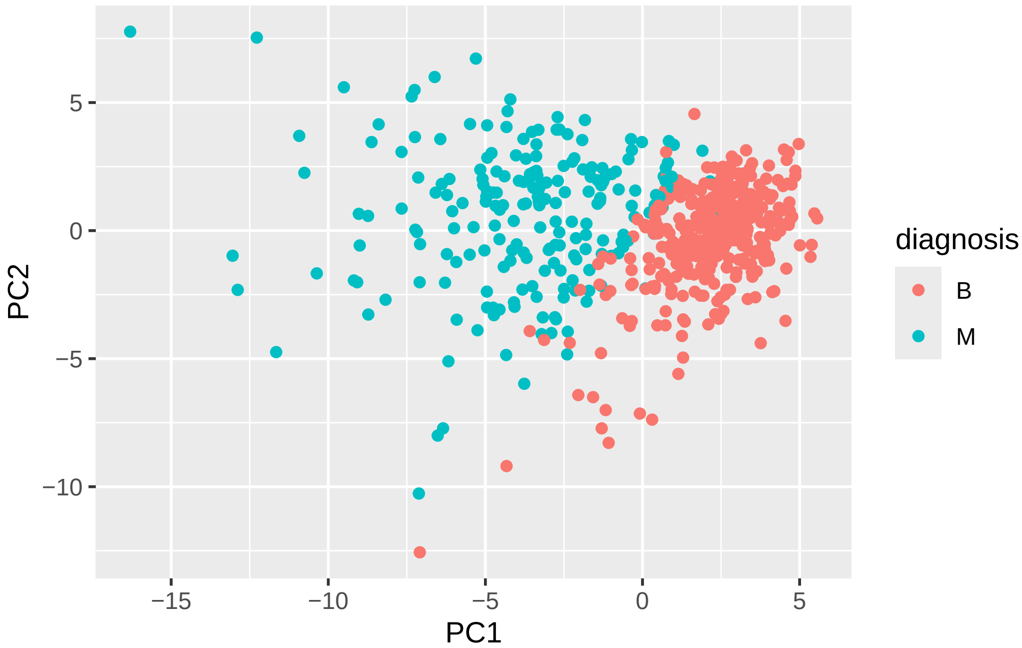
We need to scale our data before PCA with the `scale = TRUE` argument to `prcomp()`.

```r
wisc.pr <- prcomp(wisc.data, scale = T)
summary(wisc.pr)
```

```
Importance of components:
                          PC1    PC2     PC3     PC4     PC5     PC6     PC7
Standard deviation     3.6444 2.3857 1.67867 1.40735 1.28403 1.09880 0.82172
Proportion of Variance 0.4427 0.1897 0.09393 0.06602 0.05496 0.04025 0.02251
Cumulative Proportion  0.4427 0.6324 0.72636 0.79239 0.84734 0.88759 0.91010
                           PC8    PC9    PC10   PC11    PC12    PC13    PC14
Standard deviation     0.69037 0.6457 0.59219 0.5421 0.51104 0.49128 0.39624
Proportion of Variance 0.01589 0.0139 0.01169 0.0098 0.00871 0.00805 0.00523
Cumulative Proportion  0.92598 0.9399 0.95157 0.9614 0.97007 0.97812 0.98335
                          PC15    PC16    PC17    PC18    PC19    PC20   PC21
Standard deviation     0.30681 0.28260 0.24372 0.22939 0.22244 0.17652 0.1731
Proportion of Variance 0.00314 0.00266 0.00198 0.00175 0.00165 0.00104 0.0010
Cumulative Proportion  0.98649 0.98915 0.99113 0.99288 0.99453 0.99557 0.9966
                          PC22    PC23   PC24    PC25    PC26    PC27    PC28
Standard deviation     0.16565 0.15602 0.1344 0.12442 0.09043 0.08307 0.03987
Proportion of Variance 0.00091 0.00081 0.0006 0.00052 0.00027 0.00023 0.00005
Cumulative Proportion  0.99749 0.99830 0.9989 0.99942 0.99969 0.99992 0.99997
                          PC29    PC30
Standard deviation     0.02736 0.01153
Proportion of Variance 0.00002 0.00000
Cumulative Proportion  1.00000 1.00000
```

Lets evaluate the “PC Score Plot”

```r
library(ggplot2)

ggplot(wisc.pr$x) +
  aes(PC1, PC2, col = diagnosis) +
  geom_point()
```



This plot scores each data point with specific values for PC1 and PC2. The new axis shift the graph to match the largest and second largest points of variance, PC1 and PC2 respectively. This allows for better description of the data allowing for clearer visualization of trends and patterns.

> Q4: From your results, what proportion of the original variance is captured by the first principal component (PC1)?

The first principal component (PC1) captures 44.27% of the original variance.

> Q5: How many principal components (PCs) are required to describe at least 70% of the original variance in the data?

The first 3 principal components are required to describe 70% of the original variance in the data.

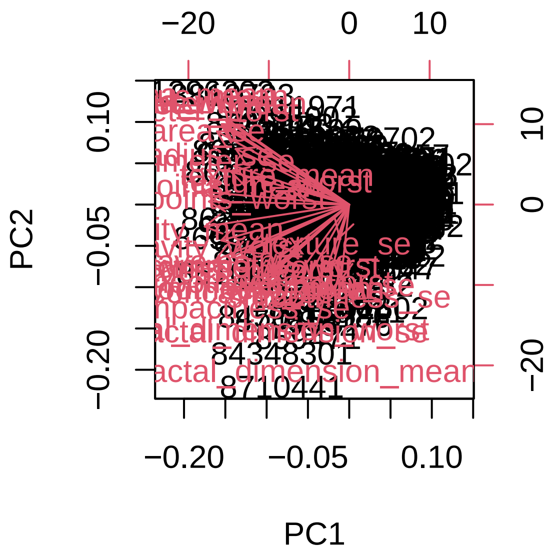
> Q6: How many principal components (PCs) are required to describe at least 90% of the original variance in the data?

The first 7 principal components are required to describe at least 90% of the original variance.

Interpreting PCA Results

A biplot is a common method to evalaute results from PCA.

```
biplot(wisc.pr)
```

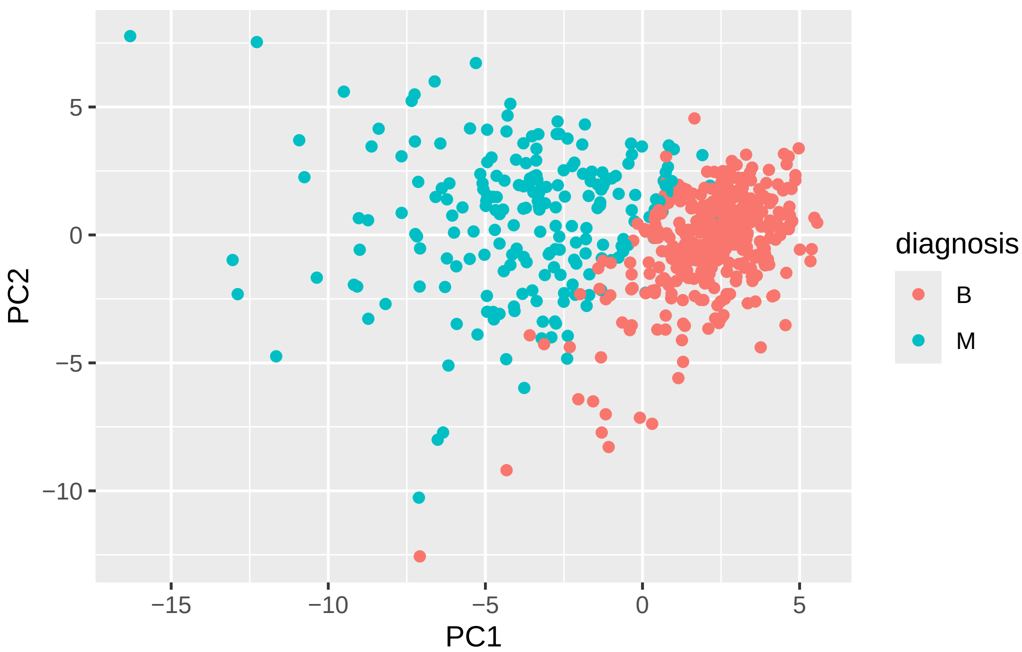


> Q7. What stands out to you about this plot? Is it easy or difficult to understand? Why?

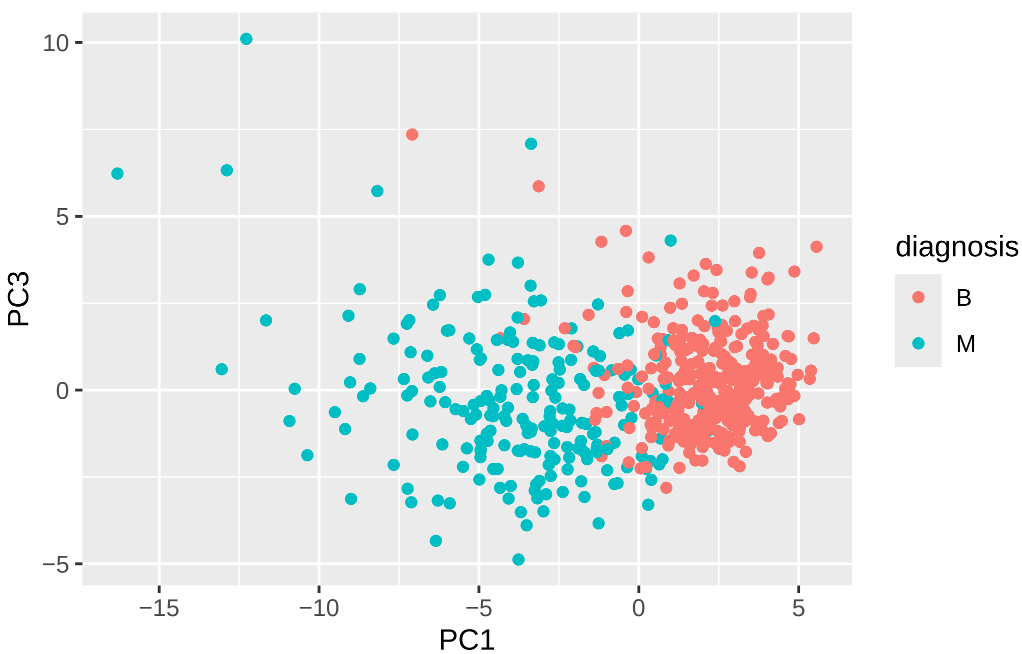
This plot is incredibly difficult to inperpret as their are multiple dimensions being plotted but everything is lumped into a minute area.

Given the difficulty to inperpret the `biplot()`, we can generate a standard scatter plot, same as the previous scatter plot. The first represents Prinicipal Components 1 and 2 while the second represents Principal Components 1 and 3.

```
ggplot(wisc.pr$x) +
  aes(PC1, PC2, col = diagnosis) +
  geom_point()
```




```r
ggplot(wisc.pr$x) +
  aes(PC1, PC3, col = diagnosis) +
  geom_point()
```



> Q8. Generate a similar plot for principal components 1 and 3. What do you notice about these plots?

It is evident that PC1 is underlying a difference between Benign and Malignant tumors, with PC2 contributing to more variance than PC3 as well.

Variance Explained

A **"scree plot"** shows how much variance each Principal Component captures. An **"elbow"** in a scree plot is important because it is the hallmark of when each PC gives diminishing return, highlighting how many PCs are a good idea to consider for appropriate analysis.

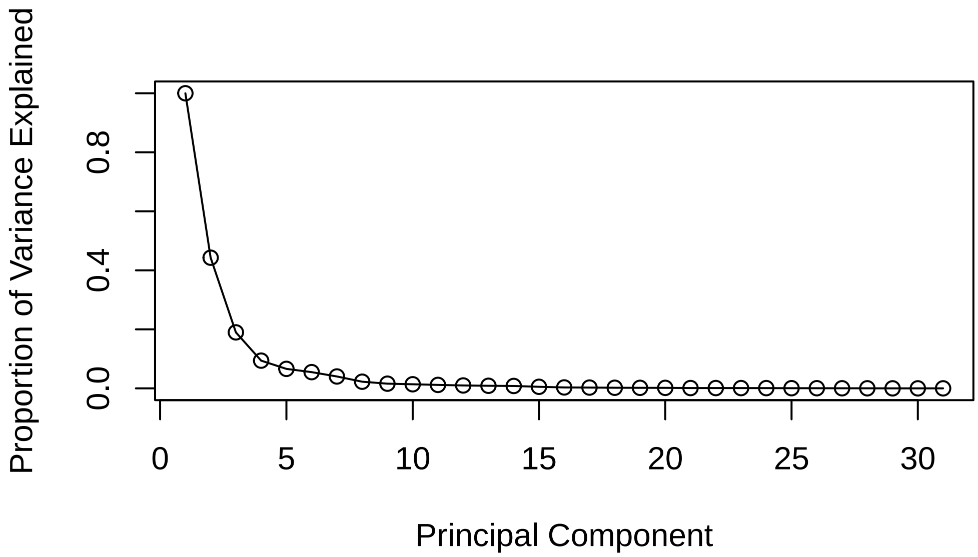
We can start calculating the variance for each principal component by squaring the sdev pf each component.

```
pr.var <- wisc.pr$sdev^2
head(pr.var)
```

```
[1] 13.281608  5.691355  2.817949  1.980640  1.648731  1.207357
```

To calculate the variance and plot it in a Skree plot, the next necessary action is to divide each principal components variance by the total variance to see what percent each PC makes of the total variance. The calculated variance percentage will be stored in the object **pve**. We can then plot the variance to build the **Skree plot**.

```
pve <- pr.var / sum(pr.var)

plot(c(1,pve), xlab = "Principal Component",
     ylab = "Proportion of Variance Explained",
     ylim = c(0, 1), type = "o")
```



Q9: For the first principal component, what is the component of the loading vector (i.e. wisc.pr$rotation[,1]) for the feature concave.points_mean? This tells us how much this original feature contributes to the first PC. Are there any features with larger contributions than this one?

```
concave_mean <- wisc.pr$rotation[,1]
concave_mean["concave.points_mean"]
```

```
concave.points_mean
         -0.2608538
```

```
threshold <- abs(concave_mean["concave.points_mean"])
concave_mean[abs(concave_mean) >= threshold]
```

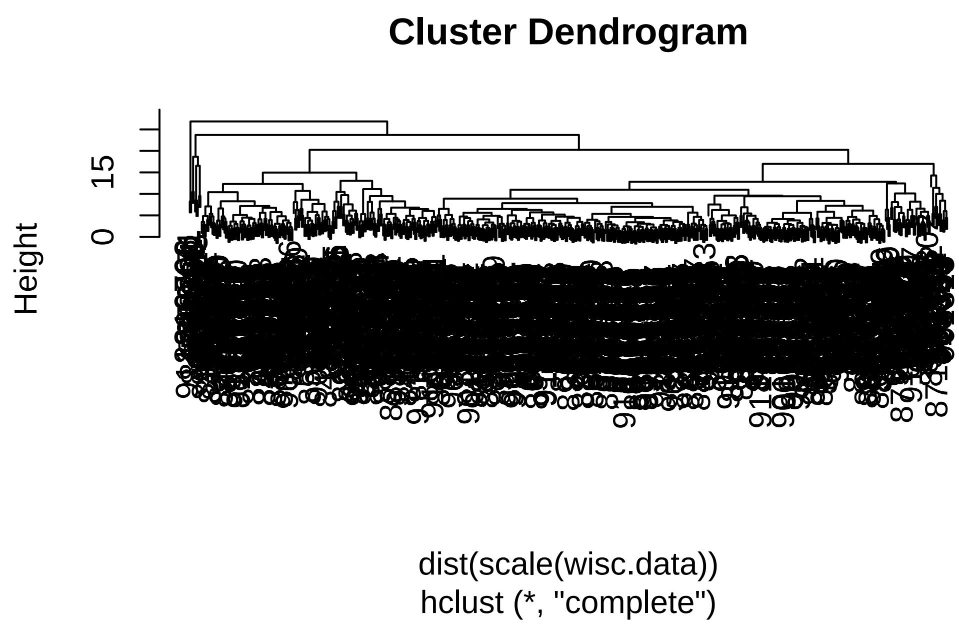
```
concave.points_mean
         -0.2608538
```

The feature "concave.points_mean" has a strong value of |0.261| which is the highest of all features.

Hierarchical Clustering

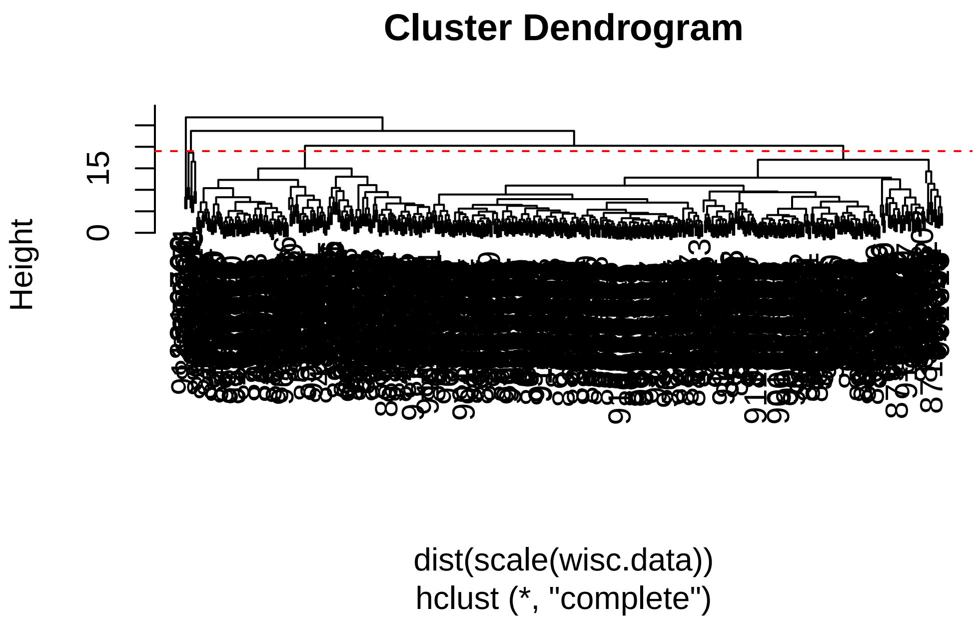
Hierarchical clustering is an important measure to determine if the original data shows obvious grouping for Malignant and Benign tumors. Hierarchical clustering is going to calculate the "distance matrix" using `hclust()` to find the distane between all pairs of observations in the data.

```
wisc.hclust <- hclust( dist( scale(wisc.data) ) )
plot(wisc.hclust)
```



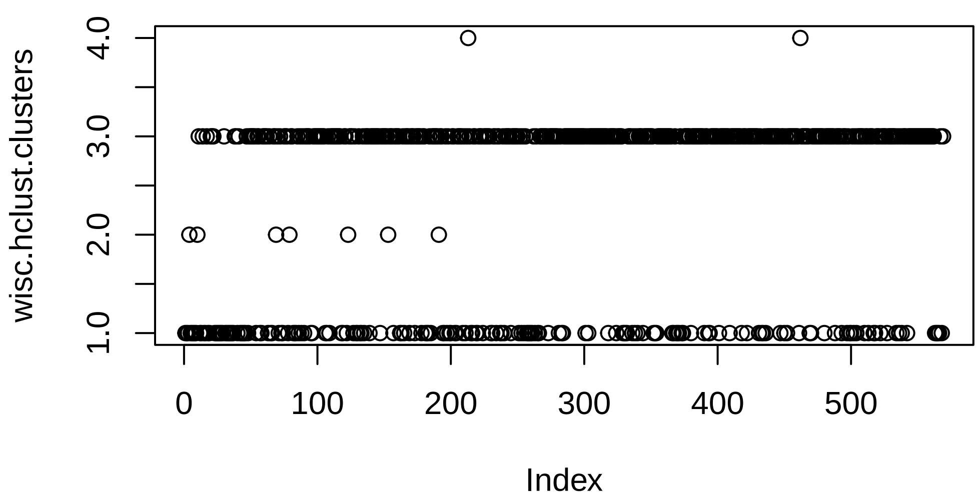
Q10: Using the plot() and abline() functions, what is the height at which the clustering model has 4 clusters?

```
plot(wisc.hclust)
abline(h = 19, col = "red", lty = 2)
```



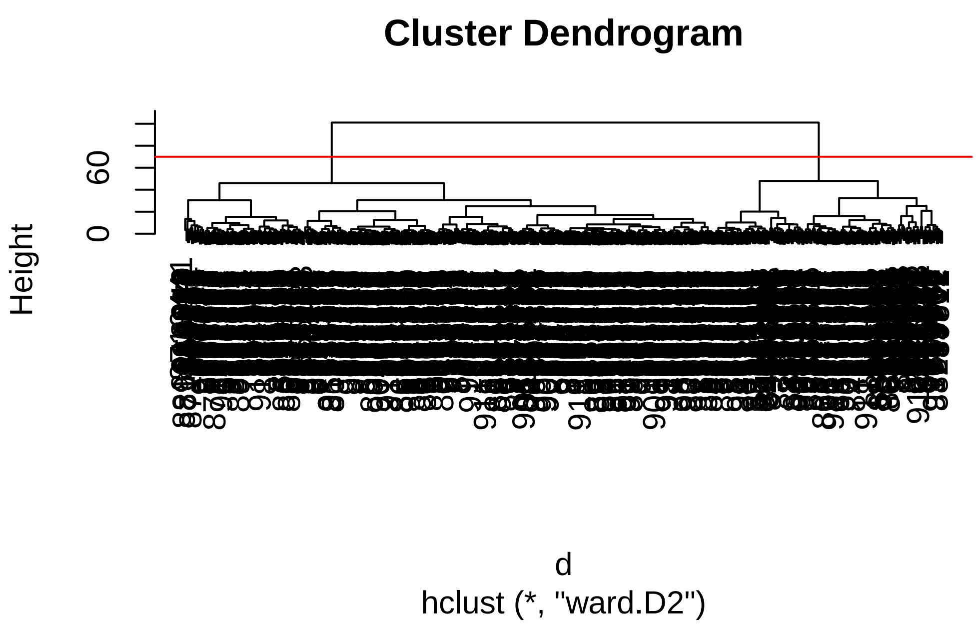
We can cut the tree at this height of 19 to create 4 clusters:

```
wisc.hclust.clusters <- cutree(wisc.hclust, h = 19)
plot(wisc.hclust.clusters)
```



Q11: Can you find a better cluster vs diagnoses match by cutting into a different number of clusters between 2 and 6? How do you judge the quality of your result in each case?

```
d <- dist( wisc.pr$x[,1:4] )
wisc.pr.hclust <- hclust(d, method = "ward.D2")
plot(wisc.pr.hclust)

abline(h=70, col = "red")
```




```
groups <- cutree(wisc.pr.hclust, h=70)
table(groups)
```

```
groups
  1   2
171 398
```

How does this clustering `groups` correspond to the expert `diagnosis`?

```
table(diagnosis)
```

```
diagnosis
  B   M 
357 212 
```

```
table(diagnosis, groups)
```

```
         groups
diagnosis   1   2
        B   6 351
        M 165  47
```

> Q12: Which method gives your favorite results for the same data.dist dataset? Explain your reasoning.

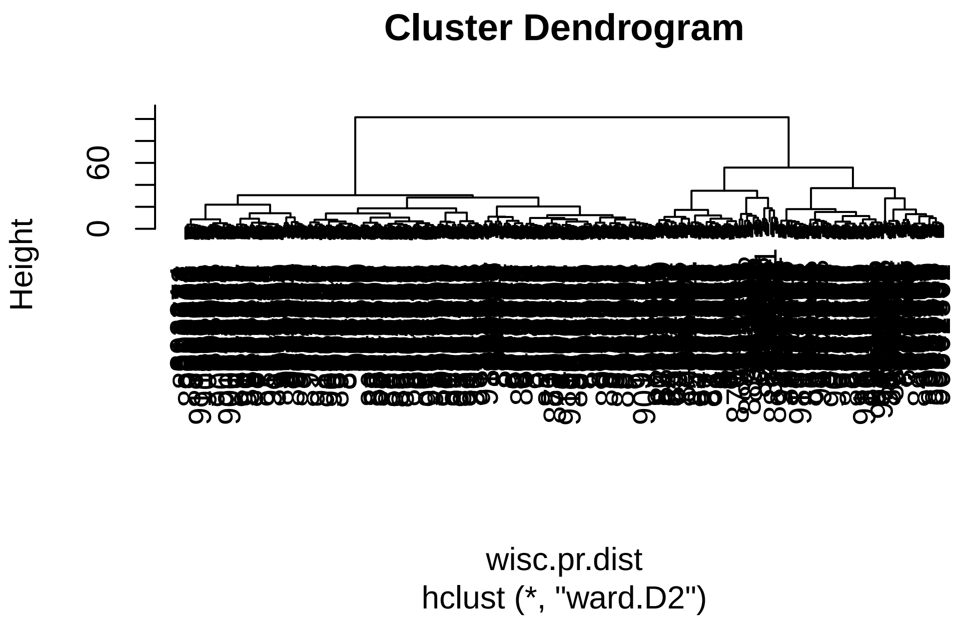
After using the "ward.D2" grouping, I prefer this one the best given that it most accurately characterizes the data and produces a clean dendrodegram

Combine Methods

Combining both PCA and Hierarchical clustering could be beneficial because PCA focuses on directions of greatest variance while clustering will focus on distance.

Using the first 7 principal components which make up 90% of the variance, we can create a hierarchical clustering model using the `ward.D2` method and assign it to the object "wisc.pr.hclust".

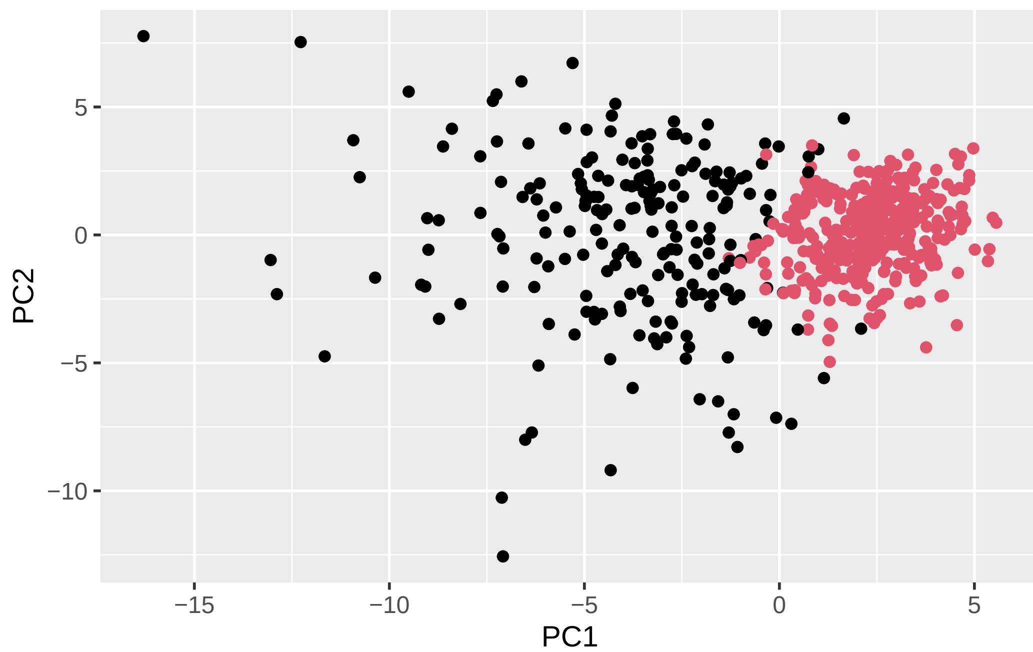
```
wisc.pr.data <- wisc.pr$x[, 1:7]
wisc.pr.dist <- dist(wisc.pr.data)
wisc.pr.hclust <- hclust(wisc.pr.dist, method = "ward.D2")

plot(wisc.pr.hclust)
```




```r
grps <- cutree(wisc.pr.hclust, k = 2)
table(grps, diagnosis)
```

```
    diagnosis
grps   B   M
   1  28 188
   2 329  24
```

```r
ggplot(wisc.pr$x) +
  aes(PC1, PC2) +
  geom_point(col=grps)
```



> Q13: How well does the newly created hclust model with two clusters separate out the two "M" and "B" diagnoses?

```r
table(grps, diagnosis)
```

```
    diagnosis
grps   B   M
   1  28 188
   2 329  24
```

Our newly created hclust model with 2 clusters for Benign and Malignant is decent, with 28 misclassified Benign tumors and 24 misclassified Malignant tumors.

> Q14: How well do the hierarchical clustering models you created in the previous sections (i.e. without first doing PCA) do in terms of separating the diagnoses? Again, use the table() function to compare the output of each model (wisc.hclust.clusters and wisc.pr.hclust.clusters) with the vector containing the actual diagnoses.

```r
table(wisc.hclust.clusters, diagnosis)
```

```
                    diagnosis
wisc.hclust.clusters   B   M
                   1  12 165
                   2   2   5
                   3 343  40
                   4   0   2
```

Without PCA their are 4 groups instead of 2, making the interpretation of the results messier and less clear. It is difficult to see exaclty how many Benign and Malignant tumors are misclassified, making the combination of PCA and H-Clustering more preferred.

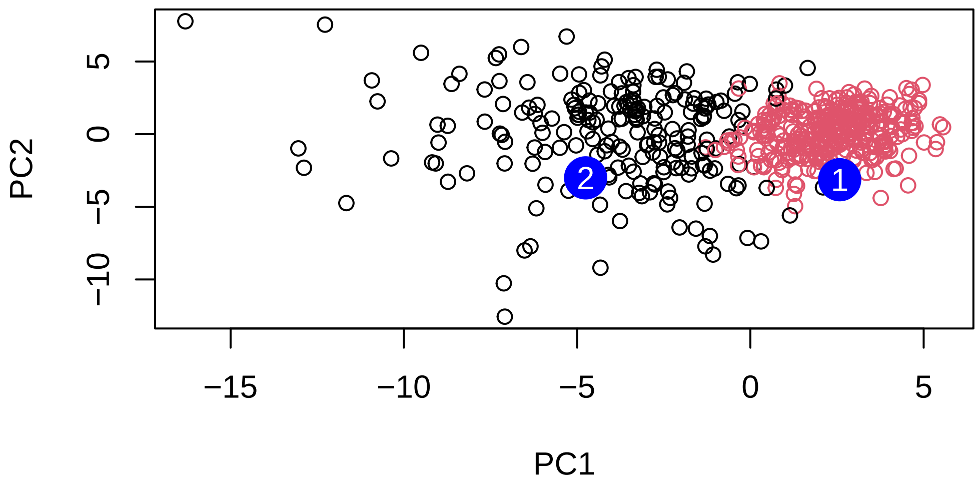
Prediction

We will use the `predict()` function that will take our PCA model from before and new cancer cell data and project that data onto our PCA space.

```
#url <- "new_samples.csv"
url <- "https://tinyurl.com/new-samples-CSV"
new <- read.csv(url)
npc <- predict(wisc.pr, newdata=new)
npc
```

```
           PC1       PC2        PC3        PC4       PC5        PC6        PC7
[1,]  2.576616 -3.135913  1.3990492 -0.7631950  2.781648 -0.8150185 -0.3959098
[2,] -4.754928 -3.009033 -0.1660946 -0.6052952 -1.140698 -1.2189945  0.8193031
            PC8       PC9       PC10      PC11      PC12      PC13     PC14
[1,] -0.2307350 0.1029569 -0.9272861 0.3411457  0.375921 0.1610764 1.187882
[2,] -0.3307423 0.5281896 -0.4855301 0.7173233 -1.185917 0.5893856 0.303029
          PC15       PC16        PC17        PC18        PC19       PC20
[1,] 0.3216974 -0.1743616 -0.07875393 -0.11207028 -0.08802955 -0.2495216
[2,] 0.1299153  0.1448061 -0.40509706  0.06565549  0.25591230 -0.4289500
           PC21       PC22       PC23       PC24        PC25         PC26
[1,]  0.1228233 0.09358453 0.08347651  0.1223396  0.02124121  0.078884581
[2,] -0.1224776 0.01732146 0.06316631 -0.2338618 -0.20755948 -0.009833238
             PC27        PC28         PC29         PC30
[1,]  0.220199544 -0.02946023 -0.015620933  0.005269029
[2,] -0.001134152  0.09638361  0.002795349 -0.019015820
```

```
plot(wisc.pr$x[,1:2], col=grps)
points(npc[,1], npc[,2], col="blue", pch=16, cex=3)
text(npc[,1], npc[,2], c(1,2), col="white")
```



Q16: Which of these new patients should we prioritize for follow up based on your results?

Given that a majority of clusters are centered in the Benign region, Patient 2 should be prioritized for a follow up given that they fall within the Malignant tumors.